ClipSAT : IB Diploma Mathematics: Analysis & Approaches / Applications

Unit 5.01: Topic 5: Differential & Integral Calculus

Name: _____

Q1. Consider the function fx = x^2 - 5x + 6 . What is the value of f2 ?

[ ] A: 0
[ ] B: -2
[ ] C: 4
[ ] D: 8

Q2. Consider the function fx = x^2 - 4x + 3 . The graph of this function is shown below. Find the y-intercept of the graph.

[ ] A: 3
[ ] B: -1
[ ] C: 0
[ ] D: -3

Q3. Consider the function fx = x^2 - 2x + 1 . Determine the y-coordinate of the vertex of the graph of y = fx .

[ ] A: 1
[ ] B: 0
[ ] C: -1
[ ] D: 2

Q4. Consider the function fx = 3x^2 - 2x + 5 . Find the value of f'1 .

[ ] A: 4
[ ] B: 5
[ ] C: 6
[ ] D: 7

Q5. The graph of the function fx = 2x^2 - 3x + 5 is shown below. Determine the value of f'2.

[ ] A: 7
[ ] B: 5
[ ] C: 9
[ ] D: 2

Q6. Consider the function fx = x^2 - 6x + 5 . Find the x-coordinate of the vertex of the parabola represented by y = fx .

[ ] A: 1
[ ] B: 3
[ ] C: -1
[ ] D: -3

Q7. Find displaystyleint x^{2}+4x^{3},dx .

[ ] A: dfrac{1}{3}x^{3}+x^{4}+C
[ ] B: dfrac{1}{4}x^{4}+dfrac{4}{5}x^{5}+C
[ ] C: 3x^{3}+16x^{4}+C
[ ] D: x^{2}+4x^{3}+C

Q8. Find displaystyleint 5x^{2}+4x,dx .

[ ] A: dfrac{5}{3}x^{3}+2x^{2}+C
[ ] B: dfrac{5}{4}x^{4}+dfrac{4}{3}x^{3}+C
[ ] C: 15x^{3}+8x^{2}+C
[ ] D: 5x^{2}+4x+C